



Habib University
shaping futures

WATER : SCIENCE, SOCIETY & POLICY

ENVS/SDP 251

Spring Semester 2025

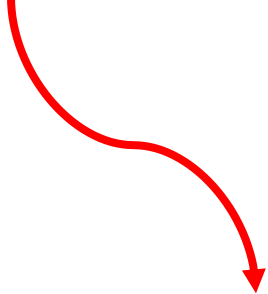
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Office hour: Thurs (9:45-11:45 PM)

Office: C-103

Email: usman.salahuddin@sse.habib.edu.pk

BERNOULLI's EQUATION



**FLUID
DYNAMICS**



**FLUID
STATICS**



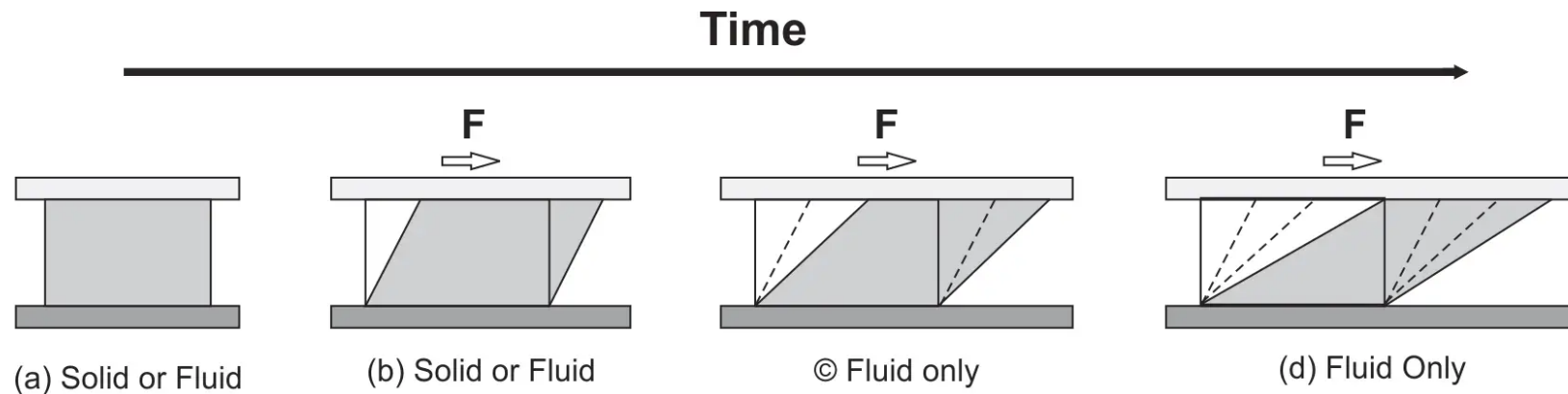
FLUID

WHAT IS A FLUID?

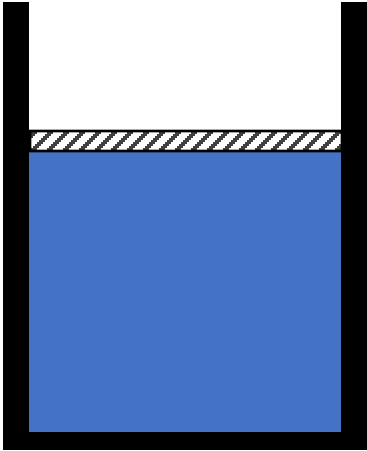
- Anything that can flow (also described as the ability to take on the shape of the container).

WHAT IS A FLUID?

- Anything that can flow (also described as the ability to take on the shape of the container).
- A fluid is a material that continuously deforms (flows) under an applied shear stress, or external force.



FLUID STATICS

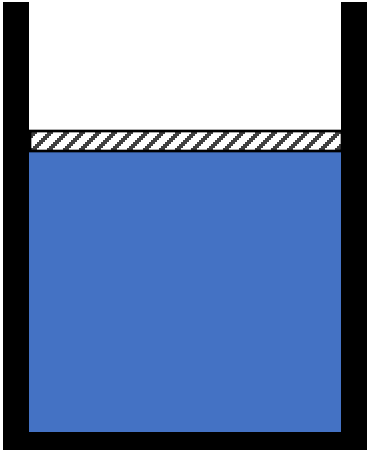


Density

$$\rho = \frac{\text{mass}}{\text{volume}}$$

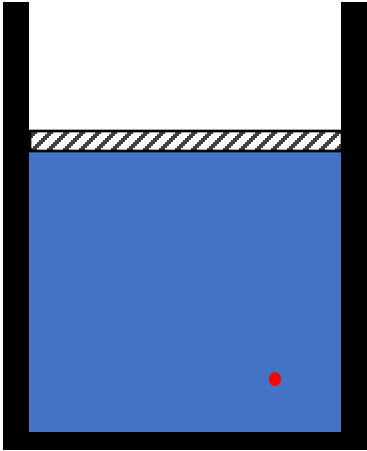
$$\rho_w = 1000 \frac{\text{kg}}{\text{m}^3}$$

FLUID STATICS



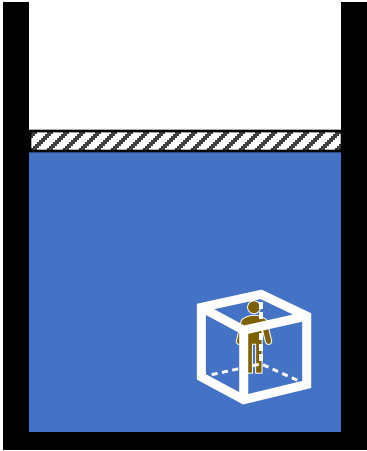
Pressure

FLUID STATICS



Pressure

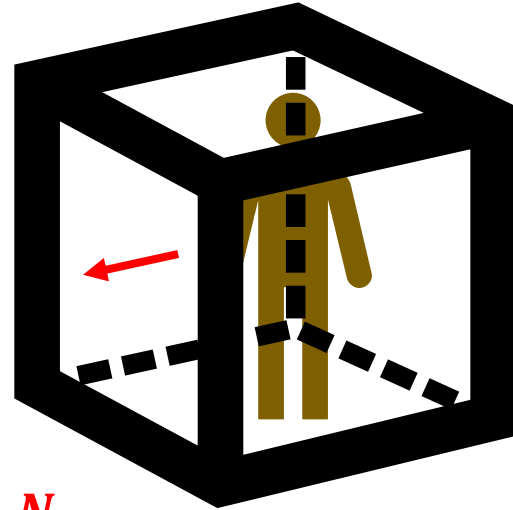
FLUID STATICS



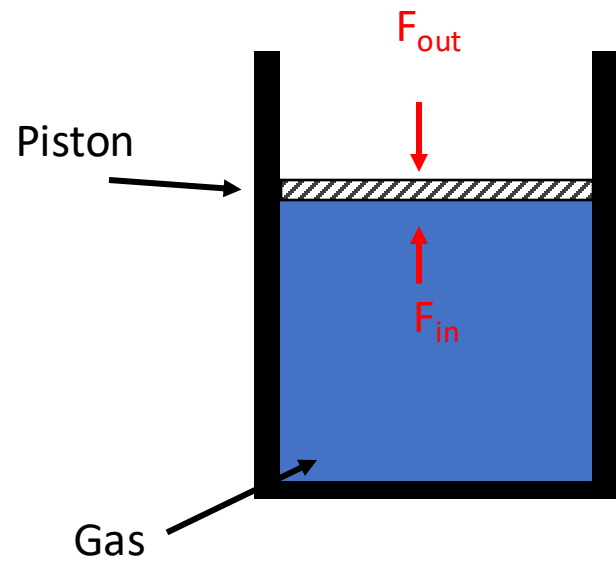
Pressure

$$P = \frac{F}{A}$$

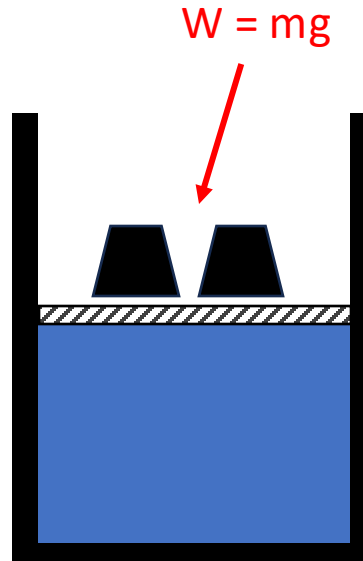
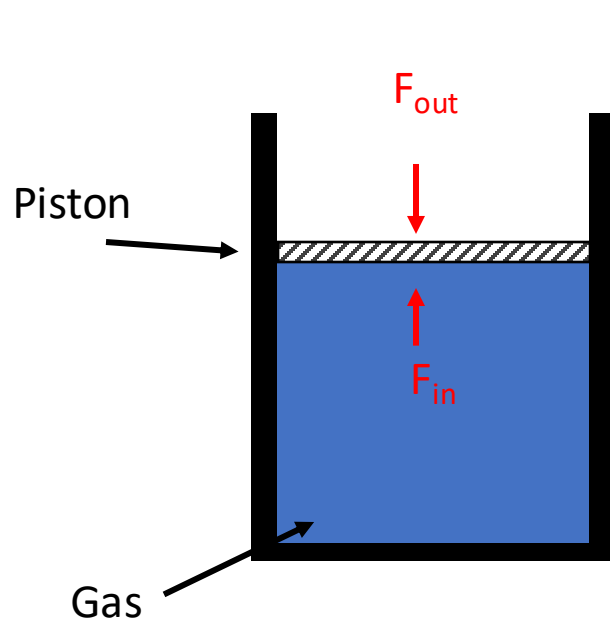
$$\text{Pascal} = \frac{N}{m^2}$$



FLUID STATICS



FLUID STATICS



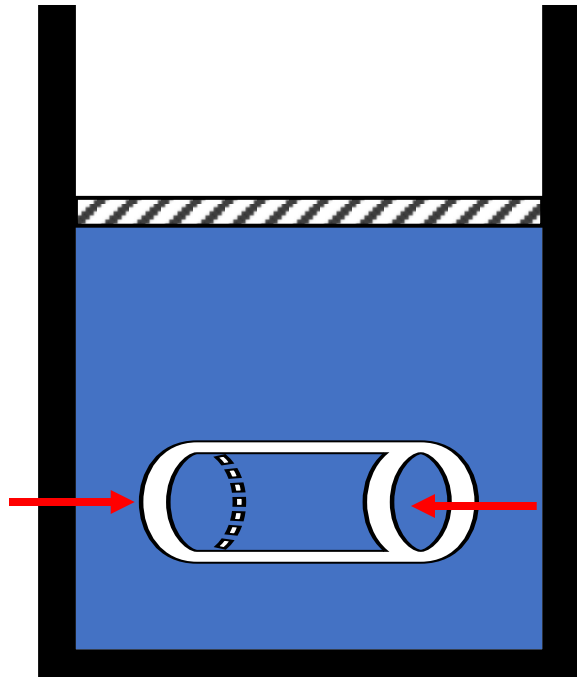
$$P = P_0 + \frac{mg}{A}$$

Atmospheric pressure

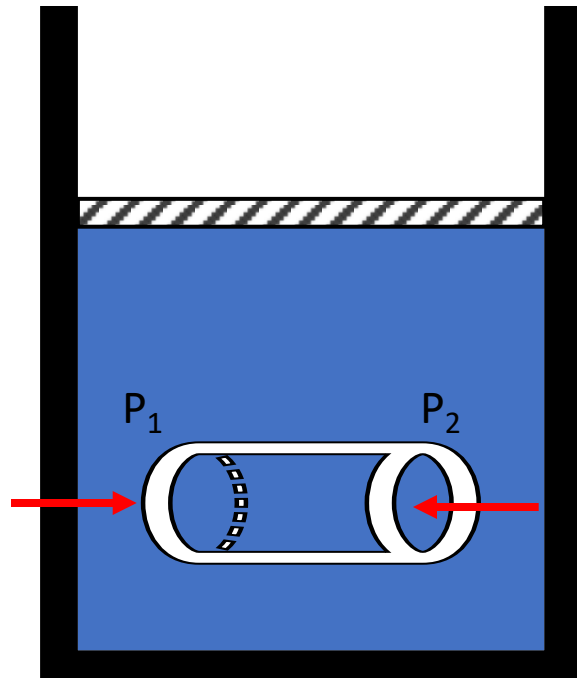
Absolute pressure

Gauge pressure

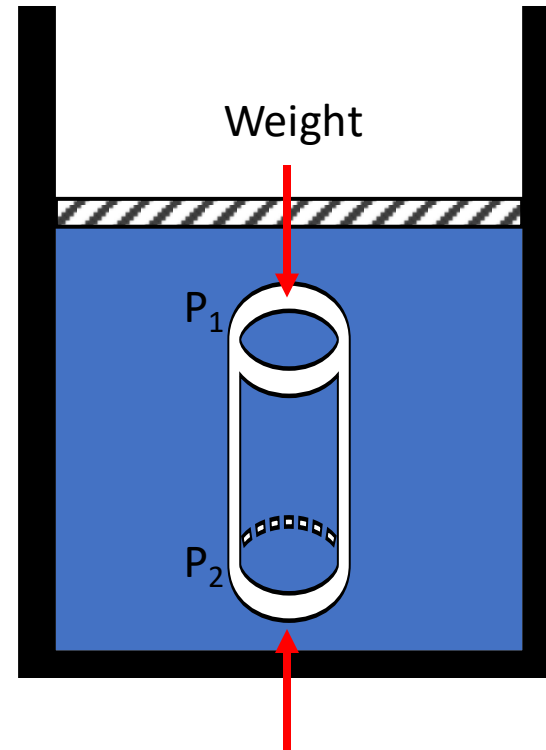
FLUID STATICS



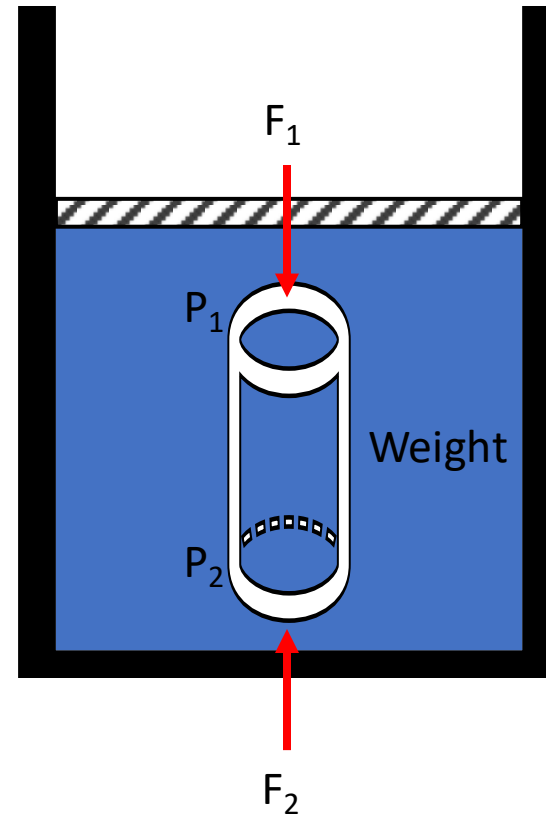
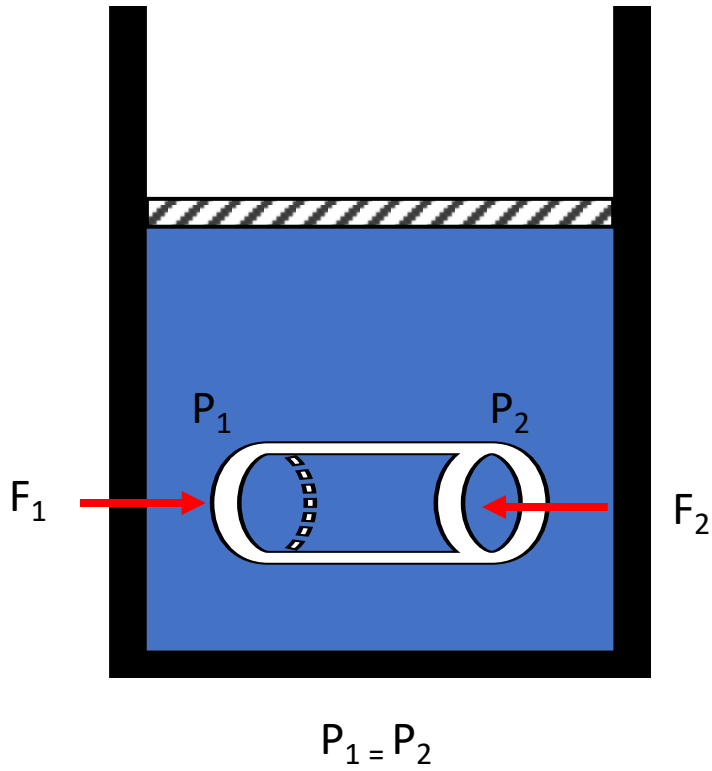
FLUID STATICS



$$P_1 = P_2$$



FLUID STATICS



$$F_2 - F_1 = W$$

$$P_2 A - P_1 A = A(h_2 - h_1)\rho g$$

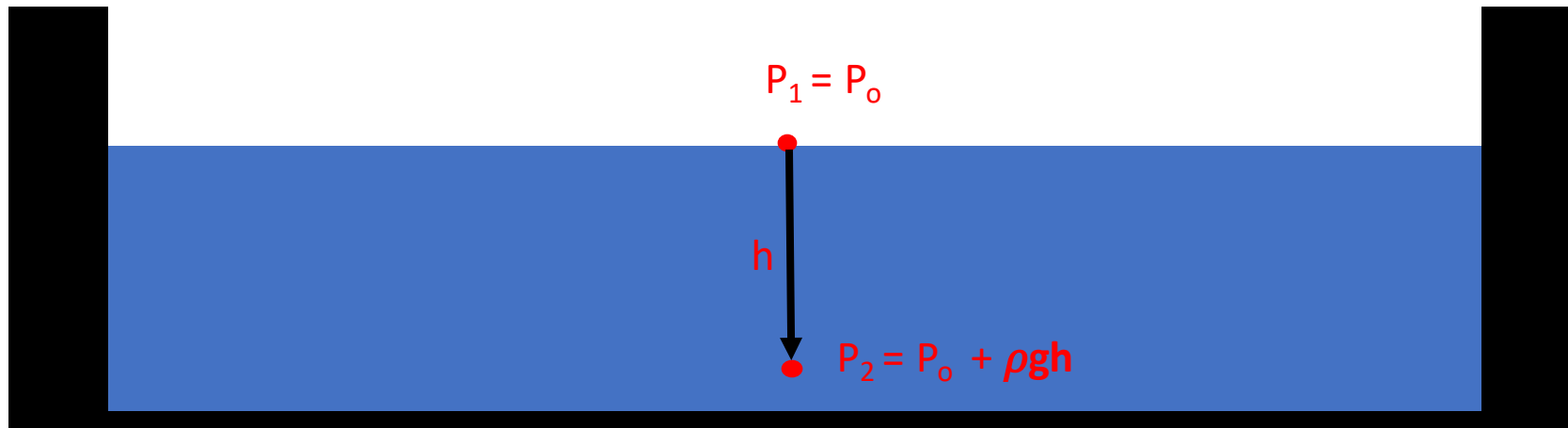
$$P_2 - P_1 = \rho g h$$

Under Water

$$P_2 - P_1 = \rho gh$$

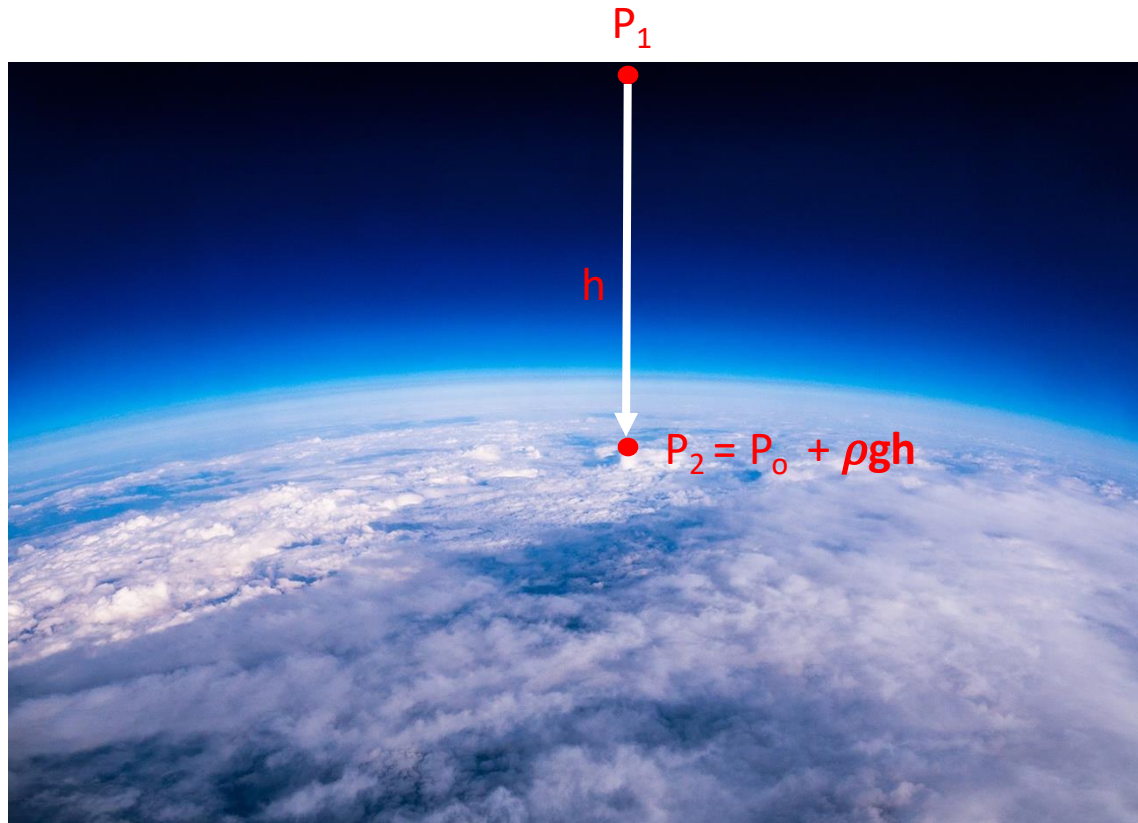


$$P_2 = P_1 + \rho gh$$



A big pool of water

Earth is a big pool!



$$P_1 = 0 \text{ (in space)}$$

$$P_2 = 0 + \rho gh$$

$$P_2 = P_0 = \rho gh$$

**Thank you for you
time and attention**

